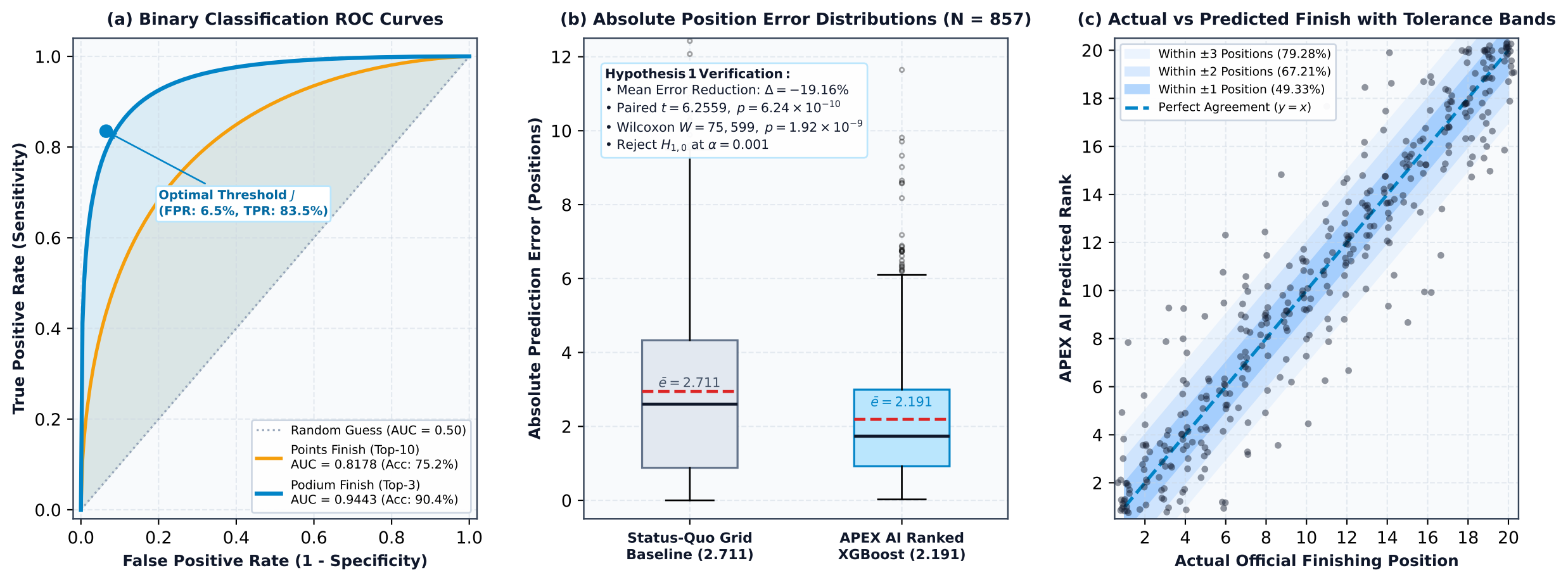

(a) Binary Classification ROC Curves
True Positive Rate (Sensitivity)
False Positive Rate (1 - Specificity)
Optimal Threshold J
(FPR: 6.5%, TPR: 83.5%)
Random Guess (AUC = 0.50)
Points Finish (Top-10)
AUC = 0.8178 (Acc: 75.2%)
Podium Finish (Top-3)
AUC = 0.9443 (Acc: 90.4%)
(b) Absolute Position Error Distributions (N = 857)
Absolute Prediction Error (Positions)
Hypothesis 1 Verification :
• Mean Error Reduction: Δ = −19.16%
• Paired t = 6.2559, p = 6.24 × 10^−10
• Wilcoxon W = 75, 599, p = 1.92 × 10^−9
• Reject H1, 0 at α = 0.001
ē = 2.711
ē = 2.191
Status-Quo Grid
Baseline (2.711)
APEX AI Ranked
XGBoost (2.191)
(c) Actual vs Predicted Finish with Tolerance Bands
APEX AI Predicted Rank
Actual Official Finishing Position
Within ±3 Positions (79.28%)
Within ±2 Positions (67.21%)
Within ±1 Position (49.33%)
Perfect Agreement (y = x)